M3T1 BACHY105 · 9 Hours · Conductors, Semiconductors, YBCO Superconductors, Magnetism, Nanomaterials & Coolants

Functional Materials & Liquid Coolants

Complete editorial study notes covering electrical resistivity factors, energy band theory, hard-drawn vs. annealed copper, ACSR overhead cables, complete typewise semiconductor classification, Meissner effect & YBCO (1:2:3) high-Tc superconductors, five magnetic classes, data storage, Stöber sol–gel nanomaterials, and dielectric coolants.

M3S0 · MEMORY STRIP

Quick-Reference Equations, Constants & Material Benchmarks

Resistance, Resistivity & Conductivity	**$R = \rho (L / A)$ | $\sigma = 1 / \rho = n e \mu_e + p e \mu_h$** — $\rho(T) = \rho_o [1 + \alpha(T - T_o)]$ (Metals: $\alpha > 0$; Semiconductors: $\alpha < 0$)
Superconducting Critical Field	**$H_c(T) = H_o [1 - (T / T_c)^2]$** — Meissner State: B = 0, $\chi = -1$ | YBCO ($YBa_2Cu_3O_{7-\delta}$): T_c = 92 K (> 77 K LN_2)
Spin-Only Magnetic Moment	**$\mu_{eff} = \sqrt{[n(n + 2)]}\ \mu_B$** — 1 μ_B = eh / (4πm$_e$) = 9.274 × 10^{-24} J T^{-1} | B = $\mu_o(H + M)$ = $\mu_o\mu_r H$
Key Material Benchmarks	**Cu: 1.68×10^{-8} Ω·m | Al: 2.65×10^{-8} Ω·m** — E_g: Si = 1.12 eV · Ge = 0.67 eV · GaAs = 1.42 eV · Diamond = 5.47 eV

M3S1 · 6 FACTORS + NICHROME VS. CU

Ohm's Law, Resistance vs. Resistivity & Six Governing Factors

Ohm's Law & Resistivity vs. Resistance
Extrinsic vs. Intrinsic

Macroscopic & Microscopic Ohm's Law: V = I R and **$J = \sigma E = E / \rho$**, where R = ρ(L/A) [Ω] and ρ = R(A/L) [Ω·m].

Resistance (R, Extrinsic): Depends on specimen geometry (length L and cross-sectional area A); unit is **Ohm (Ω)**.

Electrical Resistivity (ρ, Intrinsic): Fundamental material property independent of specimen dimensions; depends only on electronic band structure, temperature, and defect state; unit is **Ohm-meter (Ω·m)**; conductivity σ = 1/ρ (S m^{-1}).

6 Factors Governing Electrical Resistivity
High-Frequency 6M Question

1. Nature of Material & Electronic Configuration: Metals with half-filled outer s-orbitals (Cu $4s^1$, Ag $5s^1$) have high free-electron density ($n \approx 8.5 \times 10^{28}$ m^{-3}) and ultra-low ρ (~10^{-8} Ω·m), whereas covalent/ionic insulators lock electrons in localized bonds (ρ > 10^{10} Ω·m).

2. Temperature (T): In **metals**, higher T increases thermal lattice vibrations (phonons), increasing electron scattering and raising ρ linearly (**α > 0, PTC**). In **semiconductors & insulators**, higher T thermally excites exponentially more carriers across E_g, causing ρ to drop sharply (**α < 0, NTC**).

3. Alloying & Chemical Impurities (Matthiessen's Rule, $\rho = \rho_{thermal} + \rho_{impurity}$): Foreign solute atoms distort periodic lattice potential and act as scattering centers. Example: **Nichrome (80% Ni, 20% Cr)** has ρ ≈ 1.10 × 10^{-6} Ω·m (~65× higher than pure Cu) and forms a protective Cr_2O_3 scale, making it ideal for electric heating coils, whereas pure Cu is used for low-loss transmission wires.

4. Cold Working / Mechanical Deformation: Rolling or drawing introduces a high density of **crystal dislocations** that scatter electrons, increasing resistivity and tensile strength (e.g., hard-drawn Cu has higher ρ than annealed Cu).

5. Annealing (Heat Treatment): Heating a cold-worked metal below its melting point followed by slow cooling relieves internal stresses, annihilates dislocations, enlarges grains, and **restores lower electrical resistivity**.

6. Mechanical Stress, Pressure & Grain Size: Fine-grained polycrystals have higher grain-boundary scattering than single crystals; elastic strain alters interatomic spacing (basis of metallic strain gauges).

M3S2 · BAND ORIGIN + 10-POINT COMPARISON TABLE

Energy Band Theory of Solids & Master Classification of Electrical Materials

Formation of Energy Bands in Solids
Pauli Exclusion Splitting

When *N* isolated atoms (~10^{23} cm^{-3}) condense into a periodic crystal lattice with interatomic spacing r_o, their outer atomic orbitals overlap. By the **Pauli Exclusion Principle**, each atomic level splits into *N* closely spaced molecular levels (~10^{-23} eV apart), forming quasi-continuous **Energy Bands** separated by forbidden regions.

Valence Band (VB): Highest occupied band containing valence electrons at 0 K.

Conduction Band (CB): Lowest unoccupied (or partially filled) band above VB where electrons move freely under an electric field.

Forbidden Energy Gap (Bandgap, $E_g = E_C - E_V$): Energy interval between the top of VB (E_V) and bottom of CB (E_C) where no allowed electronic states exist in a pure crystal.

Parameter	Conductors (Metals)	Semiconductors	Insulators (Dielectrics)
Band Structure	VB and CB **overlap** (Be, Mg, Zn) or VB is **partially filled** (Cu $4s^1$, Na $3s^1$)	Narrow forbidden gap: **E_g ≈ 0.2 – 3.0 eV** (Si: 1.12 eV, Ge: 0.67 eV, GaAs: 1.42 eV)	Wide forbidden gap: **E_g > 4 eV (5–10 eV)** (Diamond: 5.47 eV, SiO_2: ~9 eV)
Resistivity (ρ at 300 K)	Very low: **10^{-8} to 10^{-6} Ω·m**	Intermediate: **10^{-4} to 10^{6} Ω·m**	Extremely high: **10^{10} to 10^{18} Ω·m**

Parameter	Conductors (Metals)	Semiconductors	Insulators (Dielectrics)
Conductivity (σ at 300 K)	Very high: **10^6 to 10^8 S m^{-1}**	Intermediate: **10^{-6} to 10^4 S m^{-1}**	Negligible: **10^{-18} to 10^{-10} S m^{-1}**
Charge Carriers	Free electrons only	Both **electrons (in CB) and holes (in VB)**	Virtually no free carriers under normal fields
Temp. Coefficient (α)	**Positive (α > 0, PTC):** ρ increases with T	**Negative (α < 0, NTC):** ρ decreases exponentially with T	Negative (ρ decreases slightly at very high T)
Conductivity at 0 K	Remains an excellent conductor	Acts as a **perfect insulator** (VB full, CB empty)	Perfect insulator
Effect of Doping	Increases resistivity (impurity scattering)	Drastically **increases conductivity** by 10^3–10^6×	Minimal effect at low concentration
Chemical Bonding	Metallic bond (electron sea)	Covalent or polar-covalent tetrahedral network	Strong covalent or ionic bonds
Examples	Ag, Cu, Au, Al, W	Si, Ge, GaAs, InP, CdS, ZnO	Diamond, Quartz (SiO_2), Al_2O_3, PTFE, Mica, SF_6

M3S3 · CU ELECTRONIC STRUCTURE + ACSR CABLE DESIGN

Low-Resistivity Conductors (Cu, Hard-Drawn vs. Annealed, Al & ACSR) & Insulators

Why Copper is an Exceptional Conductor
[Ar] $3d^{10}$ $4s^1$ · FCC Lattice

1. Electronic Configuration ([Ar] $3d^{10}$ $4s^1$): The completely filled, spherically symmetric **$3d^{10}$ shell** effectively shields the single outer **$4s^1$ valence electron** from the nucleus, allowing $4s^1$ electrons to delocalize into a half-filled conduction band (n = 8.5 × 10^{28} m^{-3}, ρ = 1.68 × 10^{-8} Ω·m, 100% IACS standard).

2. FCC Crystal Structure: Face-Centered Cubic lattice has **12 slip systems**, giving exceptional ductility and malleability for drawing ultra-fine wires.

3. High Thermal Conductivity (401 W m^{-1} K^{-1}) & Corrosion Resistance: Rapidly dissipates Joule heat (I^2R) and resists atmospheric oxidation.

Hard-Drawn vs. Annealed Copper
3-Mark Comparison

Hard-Drawn Copper: Produced by cold-drawing through dies without heat treatment; high dislocation density gives **high tensile strength (350–450 MPa)**, low elongation (1–3%), and slightly lower conductivity (~96–97% IACS). Used for **overhead railway traction catenary wires and busbars** that require mechanical tension.

Annealed (Soft) Copper: Heated to 400–600 °C and cooled to remove dislocations; lower tensile strength (200–250 MPa), high flexibility/elongation (30–40%), and **maximum conductivity (100–101% IACS)**. Used for **transformer windings, motor coils, and flexible indoor wiring**.

Aluminum & ACSR Overhead Power Cables
Why Al Replaces Cu in Grids

Why Aluminum for Overhead Transmission: Al ([Ne] $3s^2\ 3p^1$, FCC) has ρ = 2.65 × 10^{-8} Ω·m (~61% conductivity of Cu by volume), but its density (**2.70 g cm^{-3}**) is only **30% of Copper's (8.96 g cm^{-3})**. For the same resistance per km, an Al cable has **half the mass (200% conductivity per unit weight)** and is much cheaper, reducing pylon structural loads.

ACSR (Aluminum Conductor Steel Reinforced) Architecture: Pure Al has relatively low tensile strength and sags over long spans. ACSR solves this via a **composite concentric design**: a central core of high-tensile **galvanized steel strands** (bears mechanical tension and prevents sag under wind/ice) surrounded by outer concentric layers of high-purity **EC-grade aluminum strands** (carries >95% of the AC current, aided by the *skin effect*).

Electrical Insulators (Dielectrics)
5 Functional Classes

Key Properties Required: Ultra-high volume resistivity (>10^{12} Ω·m), high **dielectric breakdown strength (kV mm^{-1})**, low dielectric loss tangent (tan δ), thermal stability, and moisture resistance.

5 Industrial Classes: (1) **Gaseous:** Dry Air (~3 kV/mm), Nitrogen, and **SF_6 (Sulfur Hexafluoride)** (electronegative gas that quenches arcs in high-voltage circuit breakers); (2) **Liquid:** Transformer mineral oil, silicone fluids, synthetic esters, and fluorocarbon coolants; (3) **Solid Polymeric:** PVC, XLPE (cross-linked polyethylene), PTFE (Teflon), and epoxy resins; (4) **Ceramic & Glass:** Porcelain suspension insulators, Al_2O_3 (alumina), and quartz; (5) **Natural Mineral:** Muscovite/phlogopite **Mica** sheets (high dielectric strength and thermal stability up to 600 °C).

M3S4 · 3 MAJOR TYPES + 8-POINT COMPARISON TABLE

Semiconductors: Complete Typewise Classification, Covalent Lattices & n-Type vs. p-Type

Type 1: Intrinsic Semiconductors
Elemental Group IV (Si, Ge)

Structure: Ultra-pure Group IV elements (**Si: Z=14, [Ne] $3s^2 3p^2$, E_g=1.12 eV**; **Ge: Z=32, [Ar] $3d^{10} 4s^2 4p^2$, E_g=0.67 eV**) where every atom forms 4 tetrahedral sp^3 covalent bonds with neighbours.

Carrier Balance & Conductivity: Thermal breaking of covalent bonds always creates electron–hole pairs in a **1:1 ratio ($n = p = n_i$)**. Intrinsic conductivity follows **$\sigma_i = n_i e (\mu_e + \mu_h) = \sigma_o \exp(-E_g / 2k_B T)$**, and Fermi level is at mid-gap: $E_F = (E_C + E_V)/2$.

Type 2: Extrinsic Semiconductors
n-Type (Donor) & p-Type (Acceptor)

(a) n-Type Semiconductor (Pentavalent Group V Doping — P, As, Sb in Si/Ge):

Four valence electrons of P form covalent bonds with 4 Si atoms; the **5th valence electron** is weakly bound (~0.045 eV in Si, ~0.01 eV in Ge) in a discrete **Donor Level (E_d) just below E_C**.

At 300 K, thermal energy ionizes $E_d \rightarrow E_C$, giving mobile **conduction electrons (majority carriers, $n \approx N_d \gg p$)** and fixed positive **P^+ donor ions**; conductivity **$\sigma_n \approx n\,e\,\mu_e \approx N_d\,e\,\mu_e$**.

(b) p-Type Semiconductor (Trivalent Group III Doping — B, Al, Ga, In in Si/Ge):

Three valence electrons of B form 3 covalent bonds with Si, leaving the **4th Si–B bond deficient by 1 electron (a positive Hole, h^+)**.

This creates an empty discrete **Acceptor Level (E_a) just above E_V (~0.045 eV in Si)**. Valence electrons from adjacent Si–Si bonds jump into E_a, creating mobile **holes in the valence band (majority carriers, $p \approx N_a \gg n$)** and fixed negative **B^- acceptor ions**; conductivity **$\sigma_p \approx p\,e\,\mu_h \approx N_a\,e\,\mu_h$**.

Type 3: Compound Semiconductors
III–V & II–VI Families

III–V Semiconductors (Group 13 + Group 15): Stoichiometric 1:1 compounds with average 4 valence electrons per atom (zincblende/wurtzite lattice) — **GaAs (E_g = 1.42 eV, direct gap, μ_e = 8500 $cm^2V^{-1}s^{-1}$ vs. 1400 in Si)**, **InP (1.35 eV)**, **GaN (3.4 eV, blue/UV LEDs)**, **GaP (2.26 eV)**, **InSb (0.17 eV, IR detectors)**.

II–VI & IV–IV Compounds: CdS (2.42 eV), **CdTe (1.45 eV)**, **ZnO (3.37 eV)**, **ZnS (3.6 eV)**, and wide-bandgap **SiC (3.2 eV)** for high-power EV inverters and optoelectronics.

Parameter	n-Type Extrinsic Semiconductor	p-Type Extrinsic Semiconductor
Dopant Group & Valency	**Group V (Pentavalent, 5 valence e^-):** P, As, Sb	**Group III (Trivalent, 3 valence e^-):** B, Al, Ga, In
Impurity Role & Ion Charge	**Donor impurity** (donates 5th e^- → becomes immobile **P^+ ion**)	**Acceptor impurity** (accepts e^- → becomes immobile **B^- ion**)
Impurity Energy Level	Discrete **Donor level (E_d)** ~0.045 eV **below E_C**	Discrete **Acceptor level (E_a)** ~0.045 eV **above E_V**
Majority Charge Carriers	**Electrons (e^-)** in the Conduction Band ($n \gg p$)	**Holes (h^+)** in the Valence Band ($p \gg n$)
Minority Charge Carriers	**Holes (h^+)** thermally generated in VB	**Electrons (e^-)** thermally generated in CB
Fermi Level Position (E_F)	Shifts **upward** between mid-gap and E_d (closer to E_C)	Shifts **downward** between mid-gap and E_a (closer to E_V)
Electrical Conductivity	$\sigma_n \approx n\,e\,\mu_e \approx N_d\,e\,\mu_e$ (higher mobility $\mu_e > \mu_h$)	$\sigma_p \approx p\,e\,\mu_h \approx N_a\,e\,\mu_h$
Overall Electrical Charge	**Electrically Neutral** (mobile e^- balanced by fixed P^+ ions)	**Electrically Neutral** (mobile h^+ balanced by fixed B^- ions)

Superconductors: Meissner Effect, Type I vs. Type II & High-TcYBCO Ceramic

Superconductivity, T_c, H_c & Meissner Effect
ρ = 0 & Perfect Diamagnetism

Zero DC Resistivity: Discovered by Kamerlingh Onnes (1911) in Hg at 4.2 K. Below a material-specific **Critical Temperature (T_c)**, DC electrical resistivity drops abruptly to **exact zero (ρ = 0)**; persistent currents in superconducting rings flow for years without decay.

BCS Theory (Cooper Pairs): Below T_c, electrons overcome Coulomb repulsion via attractive **electron–phonon–electron interactions**, binding opposite-spin/momentum electrons into bosonic **Cooper Pairs ($e^-\uparrow, e^-\downarrow$)** that condense into a coherent quantum state and move without lattice scattering.

Meissner–Ochsenfeld Effect (Perfect Diamagnetism): When cooled below T_c in a weak magnetic field ($H < H_c$), a superconductor actively **expels all magnetic flux lines from its interior** via surface screening currents: **$B = \mu_0(H + M) = 0 \Rightarrow M = -H \Rightarrow \chi = M/H = -1$**. (Distinguishes a true superconductor from a merely "perfect conductor").

Critical Magnetic Field: Superconductivity is destroyed if the applied field exceeds **$H_c(T) = H_0 [1 - (T/T_c)^2]$**.

High-T_c Ceramic Superconductor: YBCO
$YBa_2Cu_3O_{7-\delta}$ · T_c = 92 K

Formula & 1:2:3 Stoichiometry: Yttrium Barium Copper Oxide — $YBa_2Cu_3O_{7-\delta}$ (discovered by Wu & Chu, 1987; metallic ratio **Y : Ba : Cu = 1 : 2 : 3**).

Solid-State Synthesis: Stoichiometric powders of Y_2O_3, $BaCO_3$, and CuO are ground and calcined at **900–950 °C** followed by annealing in flowing O_2 at 450–500 °C: **$\frac{1}{2}Y_2O_3 + 2BaCO_3 + 3CuO + (\frac{1}{2} - \delta)/2\ O_2 \rightarrow YBa_2Cu_3O_{7-\delta} + 2CO_2\uparrow$**.

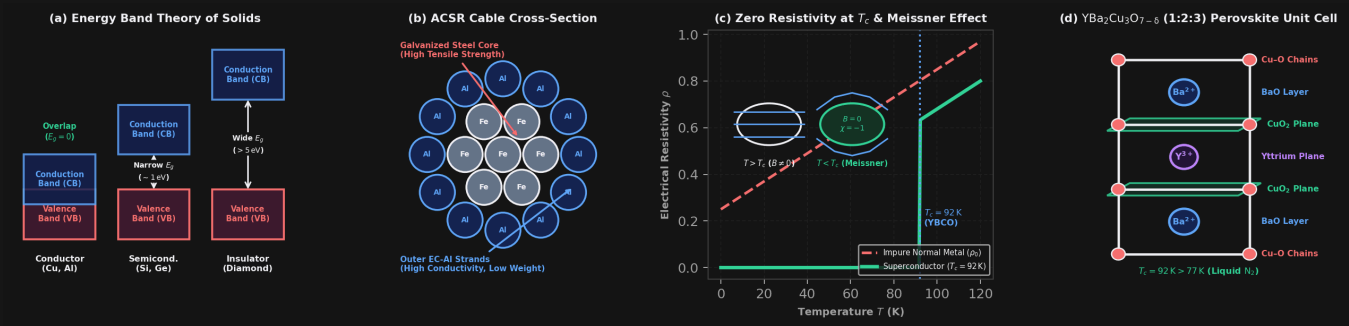
Triple-Layered Oxygen-Deficient Perovskite Structure: Orthorhombic unit cell stacked as **[CuO chains – BaO – CuO_2 plane – Y – CuO_2 plane – BaO – CuO chains]**. Mixed-valence **Cu^{2+}/Cu^{3+}** states: the 2D **CuO_2 planes** act as the superconducting Cooper-pair highways, while the 1D **Cu–O chains** act as charge reservoirs.

Oxygen Stoichiometry Control (δ): For **δ ≈ 0.07–0.15 ($YBa_2Cu_3O_{6.9}$, orthorhombic phase)**, YBCO is a superconductor with **T_c = 92 K** — the first material to superconduct **above the boiling point of inexpensive Liquid Nitrogen (77 K)** instead of costly Liquid Helium (4.2 K). For **δ > 0.6 ($YBa_2Cu_3O_6$, tetragonal phase)**, it becomes an antiferromagnetic insulator!

6 Key Applications: (1) **MRI & NMR** high-field magnets (NbTi / Nb_3Sn); (2) **Maglev trains** (Meissner levitation & flux pinning); (3) **SQUIDs** (Superconducting Quantum Interference Devices detecting 10^{-15} T brain/heart fields in MEG/MCG); (4) **Lossless power transmission cables & fault current limiters**; (5) **Particle accelerators** (CERN LHC); (6) **Superconducting qubits** in quantum computers.

Feature	Type-I (Soft) Superconductors	Type-II (Hard) Superconductors
Critical Fields	Single critical field **H_c** (very low, ~0.01–0.1 T)	Two critical fields: lower **H_{c1}** and very high upper **H_{c2}** (up to 30–100+ T)

Feature	Type-I (Soft) Superconductors	Type-II (Hard) Superconductors
Meissner Transition	Abrupt transition from Meissner state (B=0) to normal state at H_c	Complete Meissner state below H_{c1}; **Mixed (Vortex / Abrikosov) state** for $H_{c1} < H < H_{c2}$ where quantized fluxoids penetrate while R = 0 persists
Examples & Utility	Pure elements: Hg (4.15 K), Pb (7.19 K), Al (1.18 K), Sn; cannot wind high-field magnets	Alloys & cuprate ceramics: **NbTi (9.5 K), Nb_3Sn (18 K), MgB_2 (39 K), YBCO (92 K)**; used in MRI, LHC & Maglev



Visual Summary M3.1 (Slide Diagrams): *(Top Left) Energy band diagrams for Conductors, Semiconductors ($E_g \approx 1$ eV), and Insulators ($E_g > 5$ eV); (Top Right) Concentric ACSR cable cross-section with galvanized steel core and outer EC-Al strands; (Bottom Left) Meissner flux expulsion (B=0, χ=−1) and Type-I vs. Type-II phase diagrams; (Bottom Right) Triple-layered orthorhombic perovskite unit cell of $YBa_2Cu_3O_{7-\delta}$ showing CuO_2 planes and Cu–O chains (T_c = 92 K).*

M3S6 · 5-CLASS MASTER TABLE + 4 STORAGE TECHNOLOGIES

Magnetic Materials: Fundamental Dipoles, 5 Magnetic Classes & Data Storage

Origin of Atomic Magnetism & Parameters
Bohr Magneton μ_B

3 Microscopic Origins: (1) **Orbital magnetic moment (μ_L)** of electrons orbiting the nucleus; (2) **Spin magnetic moment (μ_S)** from intrinsic electron spin (s = ±½), which is the dominant contributor in 3d transition metals where crystal fields quench orbital motion; (3) Nuclear spin (negligible, ~10^{-3} μ_B).

Spin-Only Formula & Field Relations: For *n* unpaired electrons, $\mu_{eff} = \sqrt{[n(n+2)]}\ \mu_B$. Magnetic induction $B = \mu_0(H + M) = \mu_0\mu_r H$, susceptibility $\chi = M / H$, and relative permeability $\mu_r = 1 + \chi$.

Class	Dipole Alignment	Susceptibility (χ) & μ_r	Temperature Dependence	Classic Examples
1. Diamagnetic	No permanent dipoles (all e^- paired); induced dipoles oppose H	Small negative: **$\chi \approx -10^{-5}$**; $\mu_r < 1$	**Independent of T** (except superconductors where $\chi = -1$ below T_c)	Bi, Cu, Au, Ag, Si, Diamond, H_2O, NaCl
2. Paramagnetic	Random permanent dipoles (unpaired e^-); align weakly parallel to H	Small positive: **$\chi \approx +10^{-5}$ to 10^{-3}**; $\mu_r > 1$	Decreases with T: **Curie Law $\chi = C / T$** (or Curie–Weiss $\chi = C / [T - \theta]$)	O_2, Al, Pt, Ti, $CuSO_4 \cdot 5H_2O$, Fe^{3+}/Gd^{3+} salts

Class	Dipole Alignment	Susceptibility (χ) & μ_r	Temperature Dependence	Classic Examples
3. Ferromagnetic	**Parallel** spontaneous alignment of equal dipoles within domains (↑↑↑↑)	Very large positive: **$\chi \approx 10^2$ to 10^6**; $\mu_r \gg 1$	Ferromagnetic below **Curie temp T_C**; becomes paramagnetic above T_C	**Fe** (T_C=1043 K), **Co** (1388 K), **Ni** (627 K), Gd, CrO_2
4. Antiferromagnetic	**Antiparallel** alignment of **equal-magnitude** sublattice moments (↑↓↑↓ → net M = 0)	Small positive χ peaking at Néel temperature T_N	χ increases up to **Néel temp T_N**, then obeys Curie–Weiss above T_N	**MnO** (T_N=116 K), FeO, CoO, NiO, Cr, α-Fe_2O_3
5. Ferrimagnetic (Ferrites)	**Antiparallel** alignment of **unequal-magnitude** moments (↑↓↑↓ → large net M ≠ 0)	Large positive: **$\chi \approx 10^2$ to 10^4** (high electrical resistivity!)	Becomes paramagnetic above ferrimagnetic Curie temp T_C	**Magnetite (Fe_3O_4)**, Spinel ferrites MFe_2O_4 ($NiFe_2O_4$, MnZn-ferrite), YIG

Soft vs. Hard Magnetic Materials & 4 Storage Systems
Hysteresis & Curie-Point Writing

Soft Magnets (Narrow Hysteresis Loop, Low Coercivity H_c, Low Core Loss): Easily magnetized/demagnetized (Si-steel, Permalloy, MnZn-ferrites); used in **transformer cores, inductor cores, and read/write heads**.

Hard Magnets (Wide Hysteresis Loop, High Coercivity H_c & Retentivity B_r): Resist demagnetization ($Nd_2Fe_{14}B$, $SmCo_5$, Alnico, γ-Fe_2O_3, CrO_2); used in **permanent magnets and non-volatile magnetic data storage media**.

4 Magnetic Data Storage Technologies:

1. **Magnetic Hard Disk Drives (HDD):** Platter coated with high-coercivity Co–Cr–Pt thin film; inductive write head flips longitudinal/perpendicular magnetic domains (0 and 1 bits) and GMR/TMR read head senses stray flux at domain boundaries.

2. **Magnetic Tape & Stripe Cards:** Flexible polyester film coated with acicular **γ-Fe_2O_3, CrO_2, or Ba-ferrite** particles for archival backup and credit cards.

3. **Magneto-Optical (MO) Discs (Thermomagnetic Curie-Point Writing & Kerr Readout):** Uses an amorphous rare-earth transition-metal film (**TbFeCo**) with high room-temperature coercivity. To **write**, a focused laser heats a micron spot locally to its **Curie temperature ($T_C \approx 200$ °C)** where H_c drops near zero, allowing a weak bias magnet to flip the domain as it cools; to **read**, a low-power polarized laser detects the rotation of polarization via the **Magneto-Optical Kerr Effect (MOKE)**.

4. **MRAM (Magnetoresistive Random-Access Memory):** Solid-state non-volatile memory using **Magnetic Tunnel Junctions (MTJ)**: two ferromagnetic layers (pinned reference layer + free storage layer) separated by a 1 nm MgO tunnel barrier. **Parallel alignment = Low resistance (Logic 0)**; **Antiparallel alignment = High resistance (Logic 1)** via Tunnel Magnetoresistance (TMR).

M3S7 · SOL–GEL TEOS CHEMISTRY + DIELECTRIC COOLANTS

Nanomaterials (Top-Down vs. Bottom-Up Sol–Gel) & Advanced Liquid Coolants

Nanomaterials (1–100 nm) & Bulk vs. Nano Properties
Quantum Confinement & SPR

2 Drivers of Nano Behavior: (1) **Exponentially High Surface-to-Volume Ratio (S/V ∝ 1/r):** In a 2 nm particle, >50% of atoms sit on the surface with unsaturated coordination, boosting catalytic reactivity and lowering melting points; (2) **Quantum Confinement Effect:** When particle radius approaches the exciton Bohr radius, continuous bands split into discrete size-dependent levels (**smaller particle → wider E_g → blue-shifted emission**).

Classic Examples: (a) **Gold (Au):** Bulk Au is shiny yellow, chemically inert, and melts at 1064 °C; **10–20 nm Au nanoparticles** are **ruby-red** due to **Surface Plasmon Resonance (SPR) at ~520 nm**, melt at ~300–500 °C, and actively catalyze CO oxidation; (b) **CdSe Quantum Dots (2–6 nm):** Tune fluorescence from blue (2 nm) → green → red (6 nm) under a single UV lamp; (c) **Nano-TiO_2/ZnO:** Transparent UV blockers in sunscreens and self-cleaning coatings.

Top-Down (Ball Milling) vs. Bottom-Up (Sol–Gel Nano-SiO_2)
4-Step Stöber Sol–Gel Synthesis

Top-Down (High-Energy Planetary Ball Milling): Bulk coarse powder is pulverized by hard WC/ZrO_2 balls via repeated impact and shear down to 20–100 nm. Simple and scalable, but produces broad size distribution, surface defects, and mill-ball contamination.

Bottom-Up (Sol–Gel Synthesis of Monodisperse Nano-SiO_2 from TEOS): Wet-chemical self-assembly starting from molecular alkoxide precursors — **Tetraethyl Orthosilicate, $Si(OC_2H_5)_4$ (TEOS)** — in ethanol + water with NH_4OH/HCl catalyst:

1. **Hydrolysis: $Si(OR)_4 + 4H_2O \rightarrow Si(OH)_4$ (silicic acid) + 4ROH**

2. **Condensation (Water/Alcohol Elimination → Colloidal Sol): ≡Si–OH + HO–Si≡ → ≡Si–O–Si≡ (siloxane network) + H_2O**

3. **Gelation & Aging:** Sol nanoparticles cross-link into a continuous 3D wet **Alcogel / Hydrogel**.

4. **Drying & Calcination:** Evaporative drying yields a dense **Xerogel** (or supercritical drying yields an ultra-porous **Aerogel**); calcination at 400–600 °C gives ultra-pure, monodisperse **Nano-SiO_2** at low processing temperatures.

Liquid Coolants for High-Power Electronics

1000 W/cm² Heat Flux & 6 Criteria

Why Liquid Cooling: Modern AI GPUs, data-center servers, EV battery packs, and high-power lasers generate local heat fluxes approaching **100–1000 W cm^{-2}** where air cooling fails (water has ~24× higher thermal conductivity and ~3500× higher volumetric heat capacity than air).

6 Mandatory Coolant Requirements: (1) High specific heat capacity (C_p) and thermal conductivity (k); (2) Low viscosity for low pumping power; (3) Wide liquid operating range (low freezing point, high boiling point); (4) **High Flash Point (> 93 °C / 200 °F) or Non-Flammability**; (5) Chemical inertness & corrosion inhibition with Cu/Al/elastomers; (6) High dielectric breakdown strength (for direct immersion cooling) and **low Global Warming Potential (GWP) / zero ODP**.

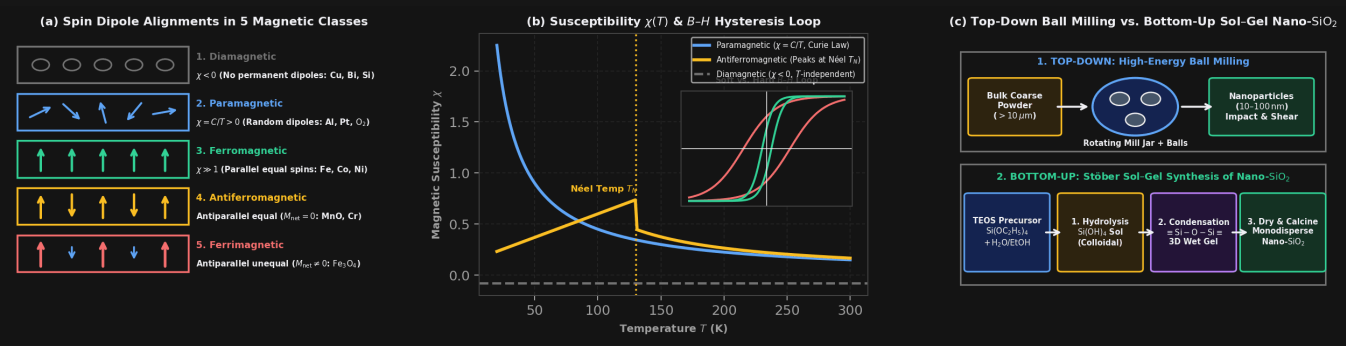
4 Industrial Coolant Classes:

1. **Aqueous DI Water + Corrosion Inhibitors:** Highest heat transfer (k = 0.60 W m^{-1} K^{-1}, C_p = 4.184 J g^{-1} K^{-1}); requires benzotriazole (BTA) for Cu passivation, molybdate/silicate for Al/steel, and biocides.

2. **Glycol–Water Antifreeze (Ethylene Glycol EG & Propylene Glycol PG):** 50:50 water–EG depresses freezing point to **−37 °C** and raises BP to 107 °C (EG has lower viscosity; PG is non-toxic/food-safe).

3. **Ultra-Low-Temperature Brines (Potassium Formate, HCOOK):** Aqueous HCOOK works down to **−50 to −70 °C** with much lower viscosity than glycols and biodegradable profile (aerospace, cold-climate radars, industrial chillers).

4. **Dielectric Fluorocarbon Fluids (Direct Two-Phase Immersion Cooling):** Electrically insulating liquids in direct contact with live motherboards: **Perfluorocarbons (PFCs, e.g., FC-72, BP 56 °C)** have high stability but **very high GWP (~9,000) and long atmospheric lifetime (3,000+ yr)**, whereas modern **Hydrofluoroethers (HFEs, e.g., Novec 7100, $C_4F_9OCH_3$)** include an ether oxygen with C–H bonds that break down in the troposphere, giving **zero ODP, short lifetime (<5 yr), and 95% lower GWP (~300)**.



Visual Summary M3.2 (Slide Diagrams): *(Top Left) Spin dipole alignments for Dia-, Para-, Ferro-, Antiferro-, and Ferrimagnetic materials; (Top Right) Magnetic storage & MTJ MRAM bit states; (Bottom Left) Top-Down Ball Milling vs. Bottom-Up Stöber Sol–Gel synthesis of Nano-SiO_2 from TEOS; (Bottom Right) Aqueous vs. Dielectric HFE/PFC immersion coolants.*

M3S8 · 5 NUMERICALS + 9 PYQ BLUEPRINTS

Complete Solved Numericals & Official VIT Question Bank Blueprints (Q1–Q9)

Solved Numerical 3.1 — Resistivity, Conductivity & Temperature Coefficient of Copper vs. Nichrome
CAT-2 / FAT Numerical · 5 Marks

Problem: A copper wire of length L = 50 m and diameter d = 1.0 mm has resistance R = 1.07 Ω at 20 °C. Calculate (a) electrical resistivity ρ, (b) conductivity σ, and (c) resistance at 120 °C given $\alpha_{Cu} = 0.00393\ °C^{-1}$.

(a) Cross-Sectional Area & Resistivity: $r = 0.5 \times 10^{-3}$ m ⇒ $A = \pi r^2 = 3.1416 \times (5 \times 10^{-4})^2 = 7.854 \times 10^{-7}\ m^2$.

$\rho = R A / L = (1.07 \times 7.854 \times 10^{-7}) / 50 = 1.681 \times 10^{-8}\ \Omega\cdot m$.

(b) Conductivity: $\sigma = 1 / \rho = 1 / (1.681 \times 10^{-8}) = 5.95 \times 10^{7}\ S\ m^{-1}$.

(c) Resistance at 120 °C (ΔT = 100 °C): $R_{120} = R_{20} [1 + \alpha\ \Delta T] = 1.07 \times [1 + 0.00393 \times 100] = 1.07 \times 1.393 =$ **1.491 Ω**.

Solved Numerical 3.2 — Intrinsic vs. Extrinsic Semiconductor Conductivity & Carrier Concentration
FAT Numerical · 5 Marks

Problem: Intrinsic silicon at 300 K has $n_i = 1.5 \times 10^{16}\ m^{-3}$, $\mu_e = 0.14\ m^2\ V^{-1}\ s^{-1}$, and $\mu_h = 0.05\ m^2\ V^{-1}\ s^{-1}$. (a) Calculate intrinsic conductivity σ_i. (b) If doped with $10^{22}\ m^{-3}$ Phosphorus atoms, find majority/minority carrier concentrations and n-type conductivity σ_n.

(a) Intrinsic Conductivity: $\sigma_i = n_i e (\mu_e + \mu_h) = (1.5 \times 10^{16})(1.6 \times 10^{-19})(0.14 + 0.05) = 4.56 \times 10^{-4}\ S\ m^{-1}$.

(b) Doped n-Type Si: Majority electrons $n \approx N_d =$ **$1.0 \times 10^{22}\ m^{-3}$**. By mass-action law, minority holes **$p = n_i^2 / n = (1.5 \times 10^{16})^2 / 10^{22} = 2.25 \times 10^{10}\ m^{-3}$**.

$\sigma_n = n e \mu_e = (1.0 \times 10^{22})(1.6 \times 10^{-19})(0.14) = 224\ S\ m^{-1}$ (conductivity increases by nearly **500,000×**!).

Solved Numerical 3.3 — Superconductor Critical Magnetic Field & Spin-Only Magnetic Moments
CAT-2 / FAT Numerical · 5 Marks

Problem: (a) A superconductor has $T_c = 9.2$ K and $H_0 = 0.196$ T at 0 K. Find its critical magnetic field H_c at 4.2 K. (b) Calculate spin-only magnetic moments (μ_{eff}) for Ti^{3+} (Z=22), Fe^{2+} (Z=26), Fe^{3+} (Z=26), and Cu^{2+} (Z=29).

(a) Critical Field at 4.2 K: $H_c(4.2\ K) = H_0 [1 - (T / T_c)^2] = 0.196 \times [1 - (4.2 / 9.2)^2] = 0.196 \times [1 - 0.2084] = 0.155$ T.

(b) Spin-Only Moments $\mu_{eff} = \sqrt{[n(n+2)]}\ \mu_B$:

Ti^{3+} ($3d^1$, n = 1): √(1×3) = **1.732 μ_B** | **Fe^{2+} ($3d^6$, n = 4):** √(4×6) = **4.899 μ_B**

Fe^{3+} ($3d^5$, n = 5): √(5×7) = **5.916 μ_B** | **Cu^{2+} ($3d^9$, n = 1):** √(1×3) = **1.732 μ_B**.